Reciprocity and Democratic Accountability*

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Abstract

We introduce reciprocity concerns in a political agency model with both symmetric learning about politicians' ability and moral hazard. In our framework, voters possess both prospective motives, as they aim to select competent politicians, and retrospective ones, as they reward actions seen as fair and punish actions seen as unfair. We study how institutional features affect politicians' incentives to exert effort (electoral control) and voters' removal of incompetent politicians (electoral screening). We show that reciprocity can overturn standard results: increasing transparency of actions improves electoral control if and only if voters' reciprocity concerns are sufficiently strong, while office rents have a non-monotone effect, and reducing rents can increase electoral control. We also show that reciprocity concerns shift the competence threshold incumbents must clear to ensure reelection, generating an incumbency advantage when voters' fairness expectations are low and an incumbency disadvantage when they are high.

Keywords: Political Agency; Career Concerns; Social Preferences; Behavioral Models of Politics

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1 Introduction

The literature on electoral accountability has long debated whether elections are used by voters to select good types or to sanction poor performances. Work emphasising sanctioning motives (e.g. Ferejohn, 1986) takes a retrospective view of voting: reelection standards reflect "approval or disapproval of that which has happened before" (Key, 1966). Work emphasising selection motives (e.g. Fearon, 1999) takes a prospective view: voters use past experiences only insofar as they allow them to infer how incumbents would act in the future if reelected.

In this paper, we argue that voters' *reciprocity* concerns—a form of other-regarding preferences (Fehr and Schmidt, 2006)—offer a way to synthesise the two approaches, without dispensing with either rational updating or the possibility of expressing approval or disapproval of past actions. Reciprocity is the tendency to reward kind treatment and to punish unkind treatment, even at a material cost to oneself, and a vast empirical literature documents reciprocal motives in voters' behaviour (e.g. Finan and Schechter, 2012; Woon, 2012; Leight et al., 2020). We introduce a reciprocal voter into the leanest workhorse model of electoral accountability with both moral hazard and learning about ability—a two-period version of Ashworth (2005)'s career-concerns framework—by incorporating in the voter's utility function the utility of an incumbent seeking reelection, with a weight whose sign and magnitude depend on whether the incumbent's effort exceeds or falls short of the effort level the voter deems equitable. The voter thus rewards incumbents he perceives as kind and punishes those he perceives as unkind, while still updating beliefs about competence in a Bayesian way and weighing both concerns at the ballot box.

Accounting for voters' reciprocity overturns results from standard models. In standard career-concerns models, the incumbent exerts effort only to engage in signal jamming (Holmström, 1999): by raising performance, she inflates the voter's perception of her ability. Since observing her action directly eliminates this motive, transparency about actions reduces effort (Ashworth and Bueno de Mesquita, 2014). With reciprocity concerns, a new channel emerges: when her action is observed, greater effort directly lowers the competence hurdle the incumbent must clear for reelection, a mechanism we call *kindness boosting*. Transparency shifts weight from signal jamming to kindness boosting and, when reciprocity concerns are sufficiently strong, *raises* equilibrium effort. A similar logic applies to office benefits, which both raise the value of winning reelection and scale the reward or punishment the voter attaches to the incumbent's conduct. This pushes the reelection threshold away from the prior mean of the ability distribution—where the retention decision is less responsive to effort. When the reciprocity-induced threshold shift is large enough, the second force dominates and *cutting* politicians' rents improves electoral control.

Reciprocity also affects electoral screening, as less competent politicians can be reelected when they act in a fairer way. Incumbency advantages or disadvantages emerge depending on what voters deem fair: we show that there exists a unique threshold of fairness expectations below which incumbents enjoy an electoral advantage and above which they suffer a disadvantage. Finally, since screening concerns are weighed against reciprocity concerns, reelecting incompetent politicians—or removing competent ones—can be fully consistent with rational, welfare-maximising voting.

Our work relates to the literature on electoral accountability (Ashworth, 2012) and, most closely, to career-concerns models of elections in the tradition of Ashworth (2005). We contribute a framework that captures simultaneously prospective and retrospective voting. Our synthesis differs from Duggan and Martinelli (2020), who also combine moral hazard and adverse selection, but whose retrospective discipline mechanism arises only from signalling incentives with standard preferences, whereas ours depends on voters' social preferences. We also relate to work on accountability and transparency (Prat, 2005; Fox and Van Weelden, 2012; Blumenthal, 2023, 2026), to behavioural political economy (e.g. Ashworth and Bueno de Mesquita, 2014; Little et al., 2022; Nunnari and Zapal, 2024), and to work on reciprocity in political agency: unlike Drazen and Ozbay (2019) and Dalmia et al. (2020), our focus is on reciprocal voters rather than reciprocal politicians.

2 Model

We consider a model of electoral accountability with career concerns, following Ashworth (2005). An incumbent is in office at the beginning of the game and, at the end of the first period, a representative voter chooses whether to re-elect her or replace her with a challenger. Untried politicians have a fixed, unknown *ability*, $\theta \sim \mathcal{N}\left(\overline{\theta}, \sigma_\theta^2\right)$, and uncertainty is symmetric. In each of the two periods, $t \in \{1, 2\}$, the office-holder produces a public good, which is the sum of her ability, θ; her effort, $a_t \in \mathbb{R}_+$; and a noise term $\varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2)$. When she exerts effort a_t, a politician's payoff is $w_t = B - c(a_t)$, where $B > 0$ is the benefit from office (capturing both formal compensation and ego rents) and $c(a_t)$ is the cost of effort, with $c(\cdot)$ increasing, three times continuously differentiable, strictly convex with non-decreasing second derivative, $c(0) = 0$, $c'(0) = 0$, $\lim_{a\to\infty} c'(a) = \infty$, and $c''(0) > 0$. Politicians out of office get zero. As described below, the voter may or may not observe the incumbent's first-period effort before the election. Let $\hat{a}_1$ denote the effort attributed to the incumbent by the voter: $\hat{a}_1 = a_1$ when effort is observed, and $\hat{a}_1 = a_1^\star$ when effort is unobserved, where $a_1^\star$ is the voter's equilibrium conjecture. Denoting the per-period output by y_t, the voter's utility is

$$u_1 = y_1 \tag{1}$$

$$u_2 = y_2 + \mathbb{1}_{\{\text{Keep}\}} \eta(\hat{a}_1 - a^e) w_2, \tag{2}$$

where $\mathbb{1}_{\{\text{Keep}\}}$ equals 1 if the incumbent is retained, $\eta > 0$ measures the voter's degree of *reciprocity* towards the first-period incumbent (the degree to which he internalises her second-period utility), and a^e is the effort level the voter deems *equitable*. When the incumbent's attributed effort is above a^e, the voter regards the action as fair and has a preference for rewarding her, proportional to η and to the action's kindness $(\hat{a}_1 - a^e)$; when it is below a^e, he regards the action as unfair and has a preference for sanctioning her.

The voter monitors the politician's action imperfectly: with probability $\tau \in [0, 1]$, the politician's first-period action is observed prior to the election; with complementary probability, the voter only observes the realisation of the public good, y_1. We characterise the pure-strategy perfect Bayesian equilibrium of the game; a sufficient condition for existence and uniqueness is that B is not too large, ensuring that first-order conditions characterise the incumbent's optimal effort (the precise condition is in the Appendix).

Modelling Assumptions. The transparency parameter τ captures voters' imperfect ability to disentangle a politician's contribution to the public good from economic fluctuations or other forces. Using the distinction between transparency about actions and transparency about outcomes (Prat, 2005; Fox and Van Weelden, 2012), we parametrise transparency as the probability with which the voter observes the politician's effort—reflecting, in practice, accountability journalism or administrative oversight (e.g. Ferraz and Finan, 2008).

Our specification of reciprocity follows the structure of models of reciprocal preferences (Rabin, 1993; Dufwenberg and Kirchsteiger, 2004; Cox et al., 2007): an agent attaches a weight to another player's payoff, and this weight depends—in sign and magnitude—on how kindly the other player has behaved. This distinguishes reciprocity from unconditional *altruism*: an altruistic voter would attach a constant positive weight to the politician's utility regardless of her conduct, whereas our voter rewards an incumbent who has exceeded the effort he deems equitable and is willing to sacrifice material utility to *punish* one who has fallen short of it. The voter has no benevolent feelings towards politicians *per se*; he responds in kind to their conduct, consistent with the empirical evidence cited above. Because the voter internalises the incumbent's *actual* continuation payoff from office, $w_2 = B$, the benefits from office enter his retention rule. We measure the kindness of the incumbent's action against the benchmark a^e, in the spirit of the action-based notion of kindness of Cox et al. (2007). Since effort is materially costly to the politician and benefits the voter, effort above the benchmark constitutes material kindness even though it is chosen, in equilibrium, with reelection in mind.

We take the equitable effort level a^e to be exogenous, allowing us to capture diverse fairness expectations across societies: citizens may expect effort commensurate with the rents politicians receive, or a norm of low expectations may prevail where political shirking is widespread. Our comparative statics with respect to transparency and office benefits (Proposition 2) hold for any a^e; the norm shapes, instead, the response of effort to fairness expectations (Proposition 3), screening (Proposition 4), and incumbency advantage or disadvantage (Proposition 5).

3 Equilibrium Analysis

3.1 Second Period and Election

Since the second period ends the game, a second-period office-holder exerts no effort: $a_2^\star = 0$. Denoting by $\tilde{\theta}$ the posterior mean of the voter's belief about the incumbent's ability, the voter reelects the incumbent if and only if

$$\text{EU(Retain)} = \tilde{\theta} + \eta(\hat{a}_1 - a^e)B \geq \overline{\theta} = \text{EU(Replace)}. \tag{3}$$

Without reciprocity concerns, the voter would reelect if and only if $\tilde{\theta} \geq \overline{\theta}$: only the prospective aspect matters (Fearon, 1999; Ashworth, 2012). Since $\eta > 0$, the retention rule includes the reciprocity term $\eta(\hat{a}_1 - a^e)B$, where B appears because it is the utility a reelected incumbent derives from office (recall $a_2^\star = 0$): this is the quantity the voter internalises when rewarding fair conduct and sanctioning unfair behaviour. In pure moral hazard models (Ferejohn, 1986), the voter commits to a retrospective rule: an action threshold chosen to optimally discipline politicians. Here, the equitable effort level plays a similar role, with two differences: our voter cannot commit—his retention rule is uniquely pinned down by sequential rationality given his preferences—and the learning dimension smooths the threshold, so a politician who exceeded the equitable effort may still be voted out, and one who fell short of it may still be reelected, depending on the voter's posterior about ability.

3.2 Electoral Control

Consider the incumbent's first-period problem. With probability $1-\tau$, her action is unobserved: by rational expectations, the voter conjectures the equilibrium action $a_1^\star$ and extracts information about ability from performance, forming the posterior mean $\tilde{\theta} = \lambda(y_1 - a_1^\star) + (1-\lambda)\overline{\theta}$, with $\lambda = \sigma_\theta^2/(\sigma_\varepsilon^2 + \sigma_\theta^2)$. As in standard career-concerns models, effort is a substitute for ability and the incumbent engages in *signal jamming*: since the voter corrects performance by the conjectured effort, increases in actual effort fool him into thinking ability is high. Contrary to the standard model, however, the reelection threshold $\overline{\theta} - \eta(a_1^\star - a^e)B$ differs from $\overline{\theta}$, and this *reduces* the benefit of signal jamming: the marginal effect of effort on the reelection probability is proportional to the density of the voter's posterior at the threshold, which is highest at the prior mean. Whether the voter seeks to reward ($a_1^\star > a^e$) or punish ($a_1^\star < a^e$) the incumbent, reciprocity pushes the threshold away from the prior mean and dampens the incentive to jam the signal.

With probability τ, the action is observed: under symmetric uncertainty, observed effort carries no ability information, so the voter subtracts actual effort from performance and signal jamming is impossible. Contrary to the standard model, a novel motive emerges, which we call *kindness boosting*: since the incumbent is reelected if and only if the voter's posterior mean exceeds $\overline{\theta} - \eta(a_1 - a^e)B$, a marginal increase in (observed) effort directly lowers the reelection threshold, increasing the chance of reelection by letting less competent incumbents be retained as a reward for fairer actions.

Putting the two cases together and defining $\alpha(a_1) := \frac{-\eta(a_1 - a^e)B}{\sqrt{\lambda}\sigma_\theta}$, the incumbent chooses a_1 to maximise

$$\tau B\left[1 - \Phi\left(\alpha(a_1)\right)\right] + (1-\tau)B\left[1 - \Phi\left(\frac{-\eta(a_1^\star - a^e)B - \lambda(a_1 - a_1^\star)}{\sqrt{\lambda}\sigma_\theta}\right)\right] - c(a_1),$$

where $\Phi(\cdot)$ and $\phi(\cdot)$ are the CDF and PDF of the standard normal distribution.

Proposition 1. *There exists a unique pure-strategy equilibrium to the game. In it, the* *incumbent's first-period action $a_1^\star$ is characterised by the first-order condition*

$$\phi\left(\frac{-\eta(a_1^\star - a^e)B\sqrt{\sigma_\theta^2+\sigma_\varepsilon^2}}{\sigma_\theta^2}\right)\left[\tau\left(\frac{\eta B^2\sqrt{\sigma_\theta^2+\sigma_\varepsilon^2}}{\sigma_\theta^2}\right)+(1-\tau)\left(\frac{B}{\sqrt{\sigma_\varepsilon^2+\sigma_\theta^2}}\right)\right]-c'(a_1^\star)=0,$$

and the incumbent is reelected if and only if $\tilde{\theta} \geq \overline{\theta} - \eta(\hat{a}_1 - a^e)B$.

Increasing τ raises the likelihood that reelection is determined in the observed-action environment, shifting weight from signal jamming to kindness boosting. When reciprocity concerns are weak, transparency mainly destroys the scope for signal jamming and effort falls—the standard result (Ashworth and Bueno de Mesquita, 2014). When reciprocity concerns are sufficiently strong, the kindness-boosting motive dominates and transparency raises effort.

Proposition 2. *Suppose the equilibrium is interior ($a_1^\star > 0$) and $a_1^\star \neq a^e$:*

1. ***Transparency.*** $\frac{\partial a_1^\star}{\partial \tau} > 0$ *if and only if* $\eta > \overline{\eta} \equiv \frac{\lambda}{B}$.
2. ***Office benefits.***
   (a) *If* $\alpha(a_1^\star)^2 < 1$, *then* $\frac{\partial a_1^\star}{\partial B} > 0$.
   (b) *If* $\alpha(a_1^\star)^2 > 2$, *then* $\frac{\partial a_1^\star}{\partial B} < 0$.
   (c) *If* $1 \leq \alpha(a_1^\star)^2 < 2$, *there exists a threshold* $\overline{\tau}(\alpha(a_1^*);\eta,B,\lambda) \in [0,1)$ *such that* $\frac{\partial a_1^\star}{\partial B} > 0$ *if and only if* $\tau > \overline{\tau}(\alpha(a_1^*);\eta,B,\lambda)$.

The threshold $\overline{\eta} = \lambda/B$ has a transparent interpretation: transparency improves electoral control when reciprocity concerns exceed the informativeness of performance about ability (λ) relative to the stakes of reelection (B). Transparency of actions is thus more likely to be beneficial where performance is a noisy signal of competence—precisely where signal jamming was a weak source of discipline to begin with—and where office is valuable; since $\overline{\eta}$ does not depend on τ, effort is globally monotone in transparency. The office benefits result reflects the dual role of B: the statistic $\alpha(a_1^\star)$ measures how far (in standard deviations) the equilibrium reelection threshold lies from the prior mean, and it grows in magnitude with η. When the threshold shift is small, the standard stakes effect dominates; when it is large, which is more likely when reciprocity concerns are strong, cutting politicians' rents improves electoral control.

Finally, we examine the two parameters describing the voter's reciprocity: the fairness norm, a^e, and the strength of reciprocity concerns, η.

Proposition 3. *Suppose the equilibrium is interior ($a_1^\star > 0$) and $a_1^\star \neq a^e$:*

1. ***Fairness norms.*** $\frac{\partial a_1^\star}{\partial a^e} > 0$ *if* $a_1^\star > a^e$ *and* $\frac{\partial a_1^\star}{\partial a^e} < 0$ *if* $a_1^\star < a^e$.
2. ***Reciprocity.*** *If* $\alpha(a_1^\star)^2 < 1$*, there exists a threshold* $\hat{\tau}(\alpha(a_1^\star); \eta, B, \lambda) \in (0, 1)$ *such that* $\frac{\partial a_1^\star}{\partial \eta} > 0$ *if and only if* $\tau > \hat{\tau}(\alpha(a_1^\star); \eta, B, \lambda)$*. If* $\alpha(a_1^\star)^2 \geq 1$*, then* $\frac{\partial a_1^\star}{\partial \eta} \leq 0$ *for all* τ.

When politicians exceed the prevailing norm, more demanding expectations spur effort, pulling the reelection threshold back towards the prior mean, where retention is most responsive to behaviour. When politicians fall short of the norm, more demanding expectations are counterproductive: reelection becomes a long shot and barely responds to effort. Societies that respond to political shirking by ratcheting up their demands on politicians may thus deepen, rather than correct, the under-provision of effort. The second result answers a natural question—does voters' reciprocity improve accountability?—and the answer depends on transparency: under imperfect monitoring reciprocity only dampens signal jamming, while with (sufficiently likely) monitoring kindness boosting can more than compensate. Thus, reciprocity and transparency are complementary determinants of accountability.

3.3 Electoral Screening and Incumbency

Since conjectures are correct in equilibrium, the incumbent's equilibrium probability of reelection is $1 - \Phi\left(\alpha(a_1^\star)\right)$ regardless of the degree of monitoring, and the *ex-ante* expected ability of the second-period office-holder is $\mathbb{E}[\theta_2] = \overline{\theta} + \phi\left(\alpha(a_1^\star)\right)\sigma_\theta\sqrt{\lambda}$ (see the Appendix). Changes in equilibrium effort have two countervailing effects on $\mathbb{E}[\theta_2]$: higher effort makes it more likely that the voter retains an incumbent whose expected ability exceeds the challenger's, but it also lowers the reelection threshold, letting politicians with lower *ex-post* ability be retained as a reward for fairer actions.

Proposition 4. *The second period's incumbent expected ability is increasing in the equilibrium effort of the first period's incumbent if* $a^e > a_1^\star$ *and decreasing if* $a^e < a_1^\star$.

Without reciprocity concerns, $\mathbb{E}[\theta_2]$ would be independent of equilibrium effort, since the voter would apply a purely prospective rule (Ashworth, 2005). Combining Proposition 4 with Proposition 2 delivers the effect of institutional reforms on screening directly in terms of model primitives:

Corollary 1. *Suppose the equilibrium is interior and* $a_1^\star \neq a^e$. *The second period office-holder's expected ability is increasing in transparency,* τ, *if and only if either (i)* $\eta > \overline{\eta}$ *and* $a_1^\star < a^e$, *or (ii)* $\eta < \overline{\eta}$ *and* $a_1^\star > a^e$.

When reciprocity concerns are strong and politicians fall short of the fairness norm, transparency improves electoral control *and* screening simultaneously: there is no trade-off between the two functions of elections. When politicians exceed the norm, the same reform that disciplines incumbents also lowers the competence hurdle they must clear, worsening the expected quality of reelected politicians.

Our framework also offers a novel mechanism for incumbency advantages (e.g. Lee, 2008) and disadvantages (e.g. Klašnja and Titiunik, 2017). Without reciprocity concerns, the martingale property of beliefs implies that the *ex-ante* reelection probability is $\frac{1}{2}$ (Ashworth, 2005). With reciprocity concerns, the threshold shifts away from the prior mean; since incumbent and challenger are *ex-ante* identical, we define the incumbency advantage as $IA \equiv \frac{1}{2} - \Phi\left(\alpha(a_1^\star)\right)$.

Proposition 5. *1. There is an incumbency advantage* ($IA > 0$) *if and only if* $a_1^\star > a^e$.

2. There is an incumbency disadvantage ($IA < 0$) *if and only if* $a_1^\star < a^e$.

3. There exists a unique threshold $\hat{a}^e > 0$ such that there is an incumbency advantage if $a^e < \hat{a}^e$ and an incumbency disadvantage if $a^e > \hat{a}^e$.

When the incumbent exceeds the equitable effort level, the reciprocity term lowers the competence threshold she must clear, making her *ex-ante* more likely to be reelected than replaced; the opposite holds when she falls short of it. The third part translates this into primitives: although equilibrium effort itself responds to the fairness norm (Proposition 3), this feedback is never strong enough to overturn the direct effect of the norm, so the comparison across societies is governed by a single cut-off in fairness expectations. All else equal, societies with demanding expectations of politicians will tend to display incumbency disadvantages, as even competent incumbents who fall short of expectations are voted out; societies with low expectations will tend to display incumbency advantages, as less competent politicians retain office thanks to efforts exceeding what voters expect of them. This mechanism complements existing explanations based on the pool of candidates (e.g. Ashworth and Bueno de Mesquita, 2008), on what voters infer from incumbents' actions (e.g. Ashworth et al., 2019), or on effort requirements varying over electoral careers (e.g. Acharya et al., 2025).

Welfare. With reciprocal voters, evaluating institutions requires asking: welfare according to which criterion? The voter's *ex-ante* expected utility decomposes as

$$W = \underbrace{\overline{\theta} + a_1^\star + \mathbb{E}[\theta_2]}_{\text{material welfare}} + \underbrace{(1 - \Phi(\alpha(a_1^\star)))\,\eta(a_1^\star - a^e)B}_{\text{reciprocity utility}}.$$

The first component is the criterion of a paternalistic planner who disregards psychological payoffs; the second is the utility the voter experiences from being reciprocal. From the point of view of a paternalistic planner, reelecting an incumbent whose posterior expected ability lies below the challenger's is a screening failure, but this can be rational when taking the voter's experienced welfare into account, as the psychological benefit of rewarding kindness compensates the expected loss of competence. When politicians fall short of the fairness norm and reciprocity concerns are strong, however, the material criterion delivers an unambiguous verdict: by Propositions 2 and 4 and Corollary 1, transparency raises effort and the expected ability of the second-period office-holder together. The normative challenge raised by behavioural preferences is thus mildest exactly where transparency reform is most needed.

4 Conclusion

We introduce reciprocity concerns in a career-concerns model of political agency. Voters with reciprocity concerns are both prospective—interested in selecting competent politicians—and retrospective—inclined to reward kind actions and punish unkind ones. Accounting for reciprocity can overturn standard results. Increasing transparency of actions improves electoral control if and only if reciprocity concerns are sufficiently strong. Office rents have a non-monotone effect, and reducing rents can increase effort. Fairness norms determine whether incumbents enjoy an electoral advantage or suffer a disadvantage. Our results generate testable predictions, since both reciprocity and fairness expectations vary measurably across individuals and societies (Falk et al., 2018): all else equal, transparency-enhancing reforms should improve incumbent performance more in electorates with stronger reciprocity concerns; the relationship between the value of office and incumbent performance should be weaker, or even negative, where reciprocity concerns loom large; and incumbency advantages should be larger where voters hold less demanding expectations of politicians. The key mechanism—reciprocity shifting the competence hurdle incumbents must clear—operates through the voter's retention rule rather than through the career-concerns structure itself, and we expect analogous effects in other electoral agency frameworks. Promising extensions include endogenising the equitable effort level, allowing for heterogeneity in voters' reciprocity concerns, and embedding reciprocity in longer electoral horizons.

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Supplementary Information

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Appendix A: Proofs

Notation. Throughout, we use the following notation: $s \equiv \sqrt{\lambda}\sigma_\theta = \frac{\sigma_\theta^2}{\sqrt{\sigma_\theta^2+\sigma_\varepsilon^2}}$; $\alpha(a) \equiv \frac{-\eta(a-a^e)B}{s}$ (as defined in the main text); and $M \equiv \tau\eta B^2 + (1-\tau)\lambda B$. Note that $\frac{1}{s} = \frac{\sqrt{\sigma_\theta^2+\sigma_\varepsilon^2}}{\sigma_\theta^2}$ and $\frac{\lambda}{s} = \frac{1}{\sqrt{\sigma_\theta^2+\sigma_\varepsilon^2}}$, so that the first-order condition in Proposition 1 can be written compactly as $\Psi(a_1^\star) = 0$, where

$$\Psi(a) \equiv \phi\left(\alpha(a)\right)\frac{M}{s} - c'(a).$$

We repeatedly use the facts that $\phi'(x) = -x\phi(x)$ and that $x\phi(x)$ is bounded between $-(2\pi e)^{-1/2}$ and $(2\pi e)^{-1/2}$ (Ashworth, 2005, p. 460).

We begin by deriving a sufficient condition on B guaranteeing that (i) the incumbent's maximisation problem is strictly concave, so that its first-order condition characterises the optimal effort choice, and (ii) the equilibrium condition Ψ is strictly decreasing, so that the equilibrium is unique and the implicit function theorem delivers the comparative statics of the equilibrium effort level.

Lemma 1. *Suppose*

$$\max\{\eta B, \lambda\}^2\, B \;<\; \sqrt{2\pi e}\cdot\lambda\sigma_\theta^2\cdot c''(0). \tag{4}$$

Then, for any $\tau \in [0,1]$ and any conjecture $a_1^\star$: (i) the incumbent's maximisation problem is strictly concave; and (ii) Ψ is strictly decreasing.

Proof of Lemma 1

<u>Claim 1:</u> Under eq. (4), $F(a_1) = B\left(1-\Phi\left(\alpha(a_1)\right)\right) - c(a_1)$ is strictly concave.

<u>Proof:</u> Since $\alpha'(a_1) = -\frac{\eta B}{s}$, we have $F'(a_1) = \frac{\eta B^2}{s}\phi(\alpha(a_1)) - c'(a_1)$ and

$$F''(a_1) = \frac{\eta B^2}{s}\phi'(\alpha)\left(-\frac{\eta B}{s}\right) - c''(a_1) = \frac{\eta^2 B^3}{s^2}\,\alpha\phi(\alpha) - c''(a_1).$$

Since $\alpha\phi(\alpha) \leq (2\pi e)^{-1/2}$, $s^2 = \lambda\sigma_\theta^2$, and $\eta^2 B^3 \leq \max\{\eta B, \lambda\}^2 B$, eq. (4) implies

$$F''(a_1) < \frac{1}{\lambda\sigma_\theta^2} \cdot \sqrt{2\pi e}\,\lambda\sigma_\theta^2\, c''(0) \cdot \frac{1}{\sqrt{2\pi e}} - c''(a_1) = c''(0) - c''(a_1) \leq 0,$$

where the last inequality follows from c'' being non-decreasing. Thus, $F(a_1)$ is strictly concave.

<u>Claim 2:</u> Under eq. (4), $G(a_1) = B\left(1 - \Phi\left(z(a_1)\right)\right) - c(a_1)$, where $z(a_1) \equiv \alpha(a_1^\star) - \frac{\lambda(a_1 - a_1^\star)}{s}$, is strictly concave.

<u>Proof:</u> Since $z'(a_1) = -\frac{\lambda}{s}$, we have $G'(a_1) = \frac{\lambda B}{s}\phi(z(a_1)) - c'(a_1)$ and

$$G''(a_1) = \frac{\lambda B}{s}\phi'(z)\left(-\frac{\lambda}{s}\right) - c''(a_1) = \frac{\lambda^2 B}{s^2}\, z\phi(z) - c''(a_1).$$

Since $z\phi(z) \leq (2\pi e)^{-1/2}$ and $\lambda^2 B \leq \max\{\eta B, \lambda\}^2 B$, eq. (4) implies, by the same steps as in Claim 1, $G''(a_1) < c''(0) - c''(a_1) \leq 0$. Thus, $G(a_1)$ is strictly concave.

Since the incumbent's objective function is $\tau F(a_1) + (1-\tau)G(a_1)$ and the convex combination of two strictly concave functions is itself strictly concave, part (i) follows.

<u>Claim 3:</u> Under eq. (4), Ψ is strictly decreasing.

<u>Proof:</u> Differentiating Ψ yields

$$\Psi'(a) = \phi'(\alpha)\left(-\frac{\eta B}{s}\right)\frac{M}{s} - c''(a) = \frac{\eta BM}{s^2}\,\alpha\phi(\alpha) - c''(a).$$

Note that $\eta BM = \tau\eta^2 B^3 + (1-\tau)\eta\lambda B^2 \leq \max\{\eta^2 B^3, \eta\lambda B^2\} \leq \max\{\eta B, \lambda\}^2 B$, where the last inequality holds because $\eta\lambda B^2$ is the geometric mean of $\eta^2 B^3$ and $\lambda^2 B$. By the same steps as in Claim 1, eq. (4) then implies $\Psi'(a) < c''(0) - c''(a) \leq 0$, which proves part (ii).

□

Proof of Proposition 1

The second-period effort choice and the voter's retention rule are derived in the main text. Consider the incumbent's first-period problem. Given a conjecture $a_1^\star$ held by the voter, the incumbent's objective function is $\tau F(a_1) + (1-\tau)G(a_1)$, with F and G as defined in the proof of Lemma 1. By Lemma 1(i), this objective is strictly concave, so the incumbent's unique best response is characterised by the first-order condition

$$\tau \frac{\eta B^2}{s} \phi(\alpha(a_1)) + (1-\tau)\frac{\lambda B}{s}\phi(z(a_1)) - c'(a_1) = 0.$$

In equilibrium, the voter's conjecture is correct, $a_1 = a_1^\star$, so that $z(a_1^\star) = \alpha(a_1^\star)$ and the condition above reduces to $\Psi(a_1^\star) = 0$, which is the first-order condition stated in the proposition. For existence and uniqueness, note that $\Psi(0) = \phi(\alpha(0))\frac{M}{s} > 0$ whenever $M > 0$ (using $c'(0) = 0$), that Ψ is continuous and, by Lemma 1(ii), strictly decreasing, and that $\lim_{a\to\infty} \Psi(a) = -\infty$ since $\phi(\cdot)\frac{M}{s}$ is bounded and $\lim_{a\to\infty} c'(a) = \infty$. Hence there exists a unique $a_1^\star > 0$ solving $\Psi(a_1^\star) = 0$. Finally, since Ψ is continuously differentiable with $\Psi'(a_1^\star) < 0$, the implicit function theorem implies that $a_1^\star$ is a continuously differentiable function of each parameter $x \in \{\tau, B, \eta, a^e\}$, with $\frac{\partial a_1^\star}{\partial x}$ having the same sign as $\frac{\partial \Psi}{\partial x}$ evaluated at $a_1^\star$. We use this fact repeatedly in the proofs below.

□

Proof of Proposition 2

Throughout, we evaluate all expressions at the equilibrium effort level and write α for $\alpha(a_1^\star)$. By the implicit function theorem (see the proof of Proposition 1), $\frac{\partial a_1^\star}{\partial x}$ has the same sign as $\frac{\partial \Psi}{\partial x}$.

1. **Transparency.** Since α does not depend directly on τ, differentiating Ψ with respect

to τ yields

$$\frac{\partial \Psi}{\partial \tau} = \phi(\alpha) \frac{\eta B^2 - \lambda B}{s}.$$

Since $\phi(\cdot) > 0$, this expression is positive if and only if $\eta B^2 > \lambda B$, that is, if and only if $\eta > \frac{\lambda}{B} \equiv \overline{\eta}$.

2. **Office benefits.** Since $\frac{\partial \alpha}{\partial B} = \frac{\alpha}{B}$ and $\frac{\partial M}{\partial B} = 2\tau\eta B + (1-\tau)\lambda$, differentiating Ψ with respect to B and using $\phi'(\alpha) = -\alpha\phi(\alpha)$ yields

$$\begin{aligned}
\frac{\partial \Psi}{\partial B} &= -\alpha\phi(\alpha)\frac{\alpha}{B}\frac{M}{s} + \phi(\alpha)\frac{2\tau\eta B + (1-\tau)\lambda}{s} \\
&= \frac{\phi(\alpha)}{sB}\left[-\alpha^2\left(\tau\eta B^2 + (1-\tau)\lambda B\right) + 2\tau\eta B^2 + (1-\tau)\lambda B\right] \\
&= \frac{\phi(\alpha)}{sB}\left[\tau\eta B^2\left(2-\alpha^2\right) + (1-\tau)\lambda B\left(1-\alpha^2\right)\right].
\end{aligned}$$

(a) If $\alpha^2 < 1$, both terms in the square bracket are non-negative and at least one is strictly positive for any $\tau \in [0,1]$, so $\frac{\partial \Psi}{\partial B} > 0$.

(b) If $\alpha^2 > 2$, both terms are non-positive and at least one is strictly negative for any $\tau \in [0,1]$, so $\frac{\partial \Psi}{\partial B} < 0$.

(c) If $1 \leq \alpha^2 < 2$, the first term is non-negative and increasing in τ while the second is non-positive and increasing in τ. Rearranging, the square bracket is positive if and only if $\tau\left[\eta B\left(2-\alpha^2\right) + \lambda\left(\alpha^2-1\right)\right] > \lambda\left(\alpha^2-1\right)$, that is, if and only if

$$\tau > \overline{\tau}(\alpha; \eta, B, \lambda) \equiv \frac{\lambda\left(\alpha^2-1\right)}{\eta B\left(2-\alpha^2\right) + \lambda\left(\alpha^2-1\right)},$$

where $\overline{\tau}(\alpha; \eta, B, \lambda) \in [0,1)$, with $\overline{\tau}(\alpha; \eta, B, \lambda) = 0$ at $\alpha^2 = 1$ and $\overline{\tau}(\alpha; \eta, B, \lambda) \to 1$ as $\alpha^2 \to 2$.

□

Proof of Proposition 3

As in the proof of Proposition 2, we evaluate all expressions at the equilibrium effort level, write α for $\alpha(a_1^\star)$, and use the fact that $\frac{\partial a_1^\star}{\partial x}$ has the same sign as $\frac{\partial \Psi}{\partial x}$.

1. **Fairness norms.** Since $\frac{\partial \alpha}{\partial a^e} = \frac{\eta B}{s}$ and M does not depend on a^e, differentiating Ψ with respect to a^e yields

$$\frac{\partial \Psi}{\partial a^e} = -\alpha\phi(\alpha)\frac{\eta B}{s}\frac{M}{s} = -\alpha\,\phi(\alpha)\frac{\eta BM}{s^2}.$$

   Since $\phi(\cdot) > 0$ and $M > 0$, the sign of this expression is the sign of $-\alpha$, which is the sign of $(a_1^\star - a^e)$: positive if $a_1^\star > a^e$ and negative if $a_1^\star < a^e$.

2. **Reciprocity.** Since $\frac{\partial \alpha}{\partial \eta} = \frac{\alpha}{\eta}$ and $\frac{\partial M}{\partial \eta} = \tau B^2$, differentiating Ψ with respect to η yields

$$\begin{aligned}\frac{\partial \Psi}{\partial \eta} &= -\alpha\phi(\alpha)\frac{\alpha}{\eta}\frac{M}{s} + \phi(\alpha)\frac{\tau B^2}{s} = \frac{\phi(\alpha)}{s\eta}\left[-\alpha^2\left(\tau\eta B^2 + (1-\tau)\lambda B\right) + \tau\eta B^2\right] \\ &= \frac{\phi(\alpha)}{s\eta}\left[\tau\eta B^2\left(1-\alpha^2\right) - (1-\tau)\lambda B\alpha^2\right].\end{aligned}$$

   If $\alpha^2 \geq 1$, both terms in the square bracket are non-positive for any $\tau \in [0,1]$, so $\frac{\partial \Psi}{\partial \eta} \leq 0$. If $\alpha^2 < 1$, rearranging, the square bracket is positive if and only if

$$\tau\left[\eta B\left(1-\alpha^2\right) + \lambda\alpha^2\right] > \lambda\alpha^2 \iff \tau > \hat{\tau}(\alpha) \equiv \frac{\lambda\alpha^2}{\eta B\left(1-\alpha^2\right) + \lambda\alpha^2},$$

   where $\hat{\tau}(\alpha) \in (0,1)$ since $\alpha \neq 0$ (which holds because $a_1^\star \neq a^e$).

□

Proof of Proposition 4

For an incumbent to be reelected, the voter's posterior belief on her ability should be greater than $\overline{\theta} - \eta(a_1^\star - a^e)B$. Since conjectures are correct in equilibrium, the equilibrium probability of re-election is $1-\Phi\left(\alpha(a_1^\star)\right)$ regardless of the degree of monitoring, and the *ex-ante* expected mean of the second period office-holder's ability is given by:

$$\Phi\left(\alpha(a_1^\star)\right)\overline{\theta}+\left(1-\Phi\left(\alpha(a_1^\star)\right)\right)\left(\overline{\theta}+\frac{\phi\left(\alpha(a_1^\star)\right)}{1-\Phi\left(\alpha(a_1^\star)\right)}\sigma_\theta\sqrt{\lambda}\right),$$

where the second term uses the mean of a normal distribution truncated from below at the reelection threshold. This can be simplified as

$$\mathbb{E}[\theta_2]=\overline{\theta}+\phi\left(\alpha(a_1^\star)\right)\sigma_\theta\sqrt{\lambda}.$$

Differentiating this expression with respect to $a_1^\star$, and using $\phi'(x)=-x\phi(x)$ and $\alpha'(a_1^\star)=-\frac{\eta B}{s}$, yields

$$\frac{\partial\,\mathbb{E}[\theta_2]}{\partial a_1^\star}=\alpha(a_1^\star)\,\phi\left(\alpha(a_1^\star)\right)\frac{\eta B}{s}\,\sigma_\theta\sqrt{\lambda},$$

which is positive if and only if $\alpha(a_1^\star)>0$, that is, if and only if $(a_1^\star-a^e)$ is negative.

$\square$

Proof of Corollary 1

By the chain rule and the proof of Proposition 4,

$$\frac{\partial\,\mathbb{E}[\theta_2]}{\partial\tau}=\frac{\partial\,\mathbb{E}[\theta_2]}{\partial a_1^\star}\cdot\frac{\partial a_1^\star}{\partial\tau},$$

which is positive if and only if $\frac{\partial\,\mathbb{E}[\theta_2]}{\partial a_1^\star}$ and $\frac{\partial a_1^\star}{\partial\tau}$ have the same sign. By Proposition 4, the former is positive if and only if $a_1^\star<a^e$; by Proposition 2, the latter is positive if and only if $\eta>\overline{\eta}$. The result follows.

$\square$

Proof of Proposition 5

Recall that the equilibrium probability of re-election of the incumbent is equal to $1-\Phi\left(\alpha(a_1^\star)\right)$.

Parts 1 and 2. If $a_1^\star > a^e$, then $\alpha(a_1^\star) < 0$ and, since Φ is strictly increasing with $\Phi(0) = \frac{1}{2}$, the probability of re-election of the incumbent is strictly greater than $\frac{1}{2}$: an incumbency advantage ($IA > 0$). Conversely, if $a_1^\star < a^e$, then $\alpha(a_1^\star) > 0$ and the probability of re-election is strictly lower than $\frac{1}{2}$: an incumbency disadvantage ($IA < 0$).

Part 3. Write $a_1^\star(a^e)$ for the equilibrium effort as a function of the equitable effort level; by the proof of Proposition 1, $a_1^\star(\cdot)$ is well-defined and continuously differentiable on $[0,\infty)$. Define $H(a^e) \equiv a_1^\star(a^e) - a^e$, so that, by Parts 1 and 2, $IA > 0$ if and only if $H(a^e) > 0$. We establish three facts.

First, $H(0) > 0$: at $a^e = 0$, $\Psi(0) = \phi(\alpha(0))\frac{M}{s} > 0$ (using $c'(0) = 0$ and $M > 0$, which holds since $\eta > 0$ and $B > 0$), so $a_1^\star(0) > 0 = a^e$.

Second, $H(a^e) \to -\infty$ as $a^e \to \infty$: from the first-order condition, $c'(a_1^\star) = \phi(\alpha(a_1^\star))\frac{M}{s} \leq \phi(0)\frac{M}{s}$, so $a_1^\star(a^e) \leq (c')^{-1}\left(\phi(0)\frac{M}{s}\right) < \infty$ for all a^e; hence $H(a^e)$ is bounded above by a constant minus a^e.

Third, every zero of H is a strictly decreasing crossing: if $H(a^e) = 0$, then $a_1^\star(a^e) = a^e$, so $\alpha(a_1^\star(a^e)) = 0$ and, by the proof of Proposition 3, $\frac{\partial\Psi}{\partial a^e} = -\alpha\phi(\alpha)\frac{\eta BM}{s^2} = 0$, which implies $\frac{\partial a_1^\star}{\partial a^e} = 0$ and, thus, $H'(a^e) = -1 < 0$.

By the first two facts and continuity, H has at least one zero. Suppose, towards a contradiction, that it had two zeros $z_1 < z_2$. By the third fact, $H < 0$ immediately to the right of z_1 and $H > 0$ immediately to the left of z_2; by the intermediate value theorem, there would then exist a zero $z_3 \in (z_1, z_2)$ at which H crosses from below, i.e. with $H'(z_3) \geq 0$, contradicting the third fact. Hence H has a unique zero $\hat{a}^e > 0$, with $H > 0$ (incumbency advantage) for $a^e < \hat{a}^e$ and $H < 0$ (incumbency disadvantage) for $a^e > \hat{a}^e$.

$\square$